

S. Amir Tabatabaei H.

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EDUCATION

2022 – 2026
Istanbul, Turkey

Ph.D. in Biomedical Engineering

Koç University (KU)

- **GPA:** 3.81 / 4
- **Thesis Project:** Türkiye's National Centralized AI-Driven Decision Support System for Kidney Transplantation.
- **Supervisor:** Dr. Caner Süsal, Prof. Nurcan Tunçbağ

2019 – 2021
Tabriz, Iran

M.Sc. in Biomedical Engineering

Tabriz University of Medical Science (TBZMED)

- **GPA:** 3.9 / 4 (Ranked 1st)
- **Thesis Topic:** Building an AI predictive model for predicting bio-impedance analysis derived phase angle determinants using adults' body composition indexes.
- **Supervisors:** Dr. Mahdad Esmaili and Dr. Fariborz Rahimi

2016 – 2018
Tabriz, Iran

B.Sc. in Biomedical Engineering (Bioelectrics)

Islamic Azad University of Tabriz (IAUT)

- **GPA:** 3.5 / 4 (Ranked 3rd)
- **Thesis Topic:** Designing a smart anti-kyphosis brace with the ability to prevent waist and neck curvature.
 - ❖ Patented at the Iran Document Registration Organization
Ref. No. 85351
- **Supervisor:** Dr. Siamak Haghipour



PUBLICATIONS

Journal

- **S. Amir Tabatabaei H.**, Alexander Fichtner, Erol Demir, Nurcan Tunçbağ, Caner Süsal, Burkhard Tönshoff

"Dynamic Risk Factor Analysis Across the Post-Transplant Timeline in Pediatric Kidney Recipients: A Multi-Method Longitudinal Study."

American Journal of Kidney Transplantation, 2026 (Under review)

Journal

- **S. Amir Tabatabaei H.**, Erol Demir, Nurcan Tunçbağ, Caner Süsal, et al.

"PRE-EMPTIVE Kidney Transplantation Improves Outcomes in Elderly Patients: Propensity Score-Matched Multicenter National Study."

American Journal of Kidney Transplantation, 2026 (Under review)

Journal

- **S. Amir Tabatabaei H.**, Erol Demir, Nurcan Tunçbağ, Hamad Dheir, Caner Süsal, Aydın Türkmen

"Is Machine Learning Better Than Cox Regression in Predicting Kidney Transplant Outcomes?"

American Journal of Kidney Transplantation, 2026 (Under review)

Oral Presentation	• S. Amir Tabatabaei H., Nurcan Tunçbağ, Aydın Türkmen, Caner Süsal, et al. “Machine Learning Outperforms Traditional Statistical Methods in Predicting Kidney Transplant Outcomes: Türkiye Multicenter Cohort Study.” <i>31st International Congress of The Transplantation Society (TTS 2026) – Sydney, Australia (Link here)</i>
Oral Presentation	• S. Amir Tabatabaei H., Christian Patry, Kai Krupka, Caner Süsal, Burkhard Tönshoff, et al. “Dynamic Risk Prediction of Graft Survival in Pediatric Kidney Transplantation: Cox vs. Machine Learning.” <i>31st International Congress of The Transplantation Society (TTS 2026) – Sydney, Australia (Link here)</i>
Oral Presentation	• S. Amir Tabatabaei H., Hamad Dheir, Ülkem Çakır, et al. “Machine Learning Outperforms Traditional Methods in Predicting Kidney Transplant Outcomes: A Multicenter Cohort Study.” <i>39th European Immunogenetics and Histocompatibility Conference 2026– Edinburgh, Scotland</i>
Oral Presentation	• S. Amir Tabatabaei H., Dunya Moradi, Beren Semiz “Multi-Method Signal Decomposition Reveals Neural Network Disruption in Claustrophobia: A Multi-Scale Connectivity Analysis.” <i>In Proceedings of the 19th International BIOSIGNALS Conference, Marbella, Spain; ISBN 978-989-758-802-0; ISSN 2184-4305, SciTePress, pages 427-434, (2026), (Full Paper Here)</i>
Journal	• S. Akgul, U. Aslanhan, R. Kazamzadeh, Y. Alan, E. Demir, D. Ece Salli, S. Amir Tabatabaei H., et al. “205.6: Desensitization-resistant eplet-specific HLA antibodies.” <i>Transplantation 108(9S), September 2024, (Full Paper Here)</i>
Journal	• S. Amir Tabatabaei H., R. Kazemzadeh, BJ. Foster, E. Arpalı, C. Süsal “New Tools for Data Harmonization and Their Potential Applications in Organ Transplantation” <i>Transplantation. Epub ahead of print. PMID: 38755748, (2024), (Full Paper Here)</i>
Journal	• S. Amir Tabatabaei H., F. Rahimi, M. Esmaili, M. Khalili “Phase Angle Determinants in Patients with Cardiovascular Disease Using Machine Learning Methods.” <i>Journal of Health and Technology, (2021), (Full Paper Here)</i>
Journal	• S. Sarabi, M. Asadnejad, S. Amir Tabatabaei H., S. Rajebi “Using Artificial Intelligence for Detection of Lymphatic Disease and Investigation on Methods of Its Classifications.” <i>International Journal on "Technical and Physical Problems of Engineering", (2020), 12(43), 58-65. (Full Paper Here)</i>
Oral Presentation	• S. Amir Tabatabaei H., “AI and Data Harmonization in Organ Transplantation.” <i>Oral Presentation at the Expert Meeting on Innovative Approaches to Immunogenetics and Organ Transplantation, Koç University Hospital, Istanbul, Turkey. 2024, September.</i>

Oral Presentation	S. Amir Tabatabaei H., “New Tools for Data Harmonization in Organ Transplantation.” <i>Plenary Presentation at the 9th National Transplant Immunology and Genetics Congress, Transplant Immunology and Genetics Association (TIGED), Antalya, Turkey. 2024, April. (Congress Link Here)</i>
Oral Presentation	S. Amir Tabatabaei H., M. Esmaeili, Y. Donyatalab, F. Rahimi “The Most Effective Factors in Predicting Bioelectrical Impedance Phase Angle for Classification of Healthy and Depressed Obese Women: An Artificial Intelligence Approach.” <i>Intelligent and Fuzzy Techniques for Emerging Conditions and Digital Transformation. INFUS 2021, vol. 308. Springer, (Abstract Here)</i>
Oral Presentation	D. Moradi, R. Eyvazpour, S. Amir Tabatabaei H., Y. Donyatalab, F. Rahimi “Fuzzy Cluster Analysis of Claustrophobia Questionnaire Data in Iranian Male and Female Educated Populations.” <i>Intelligent and Fuzzy Techniques for Emerging Conditions and Digital Transformation. INFUS 2021, vol. 307. Springer, (Abstract Here)</i>

AWARDS & GRANTS

June 2026 Sydney, Australia	Congress Scientific Award, The Transplantation Society (TTS 2026) Awarded for the manuscript on "AI-based Model for Predicting Kidney Transplant Outcomes".
February 2022 Istanbul, Turkey	Research grant for Ph.D. research (Principal Investigator: Dr. C. Süsal) Funding granted from European Union's Horizon 2020 Research and Innovation Program.
June 2019 Tabriz, Iran	Research grant for M.Sc. thesis (Principal Investigator: Dr. M. Esmaeili) Funding granted from Tabriz University of Medical Science research center sources
May 2018 Tehran, Iran	Awarded Iranian government scholarship for graduate studies. Due to my 32nd place in the nationwide Iranian university entrance exam among approximately 9000 students.
June 2015 Pittsburg, USA	Gold medal at International Exhibition of Inventions in United States (INPEX) H. Abdollahzadeh, S.A. Tabatabaei H., R. Fathzadeh, M. Gholipour Project: The physician assisting robot.
October 2013 Warsaw, Poland	Gold medal at the 7th International Exhibition of Inventions in Poland (IWIS) S.A. Tabatabaei H., H. Abdollahzadeh, M. Soleymani, S. Ghadimi, S. Namazipour Project: Composite wheelchair for the physically disabled people with the ability to go up and down the stairs and becoming in the bed style.
April 2013 Geneva, Switzerland	Gold medal at the 41st International Exhibition of Inventions in Switzerland (IEIG) H. Abdollahzadeh, S.A. Tabatabaei H., S. Salimi, F. Jalilzadeh Project: The leafing Robot for physically handicapped people using brain signal processing.

PATENTS

March 2015	Smart Anti-Kyphosis Brace with the Ability to Prevent Waist and Neck Curvature. - Iran Document Registration Organization, Ref. No. 85351 - Funded by Azad University of Tabriz
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- Iran Document Registration Organization, Ref. No. 84232
- Funded by Sahand University of Technology

SKILLS

Professional	Advanced: Python, MATLAB Intermediate: SQL Basic: R Libraries: Tensorflow, Django, Keras, OpenCV, Scikit-Learn, Numpy, Pandas, etc.
Software	Tools: Proteus, Pspice, SPSS, Excel, Github Platforms: Linux, Windows
Language	Native: Azerbaijani, Persian Advanced: English (Yökdil: 87.5 / 100), Turkish Basic: German (A1)

INVOLVED PROJECTS

2024 – Current	AI-based predictive model development for short- and long-term kidney allograft loss probability after transplantation, Transplant Immunology Research Center of Excellence Lab (TIREX) , Koç University, Istanbul, Turkey.
2023 – Current	AI-based segmentation of pathological features such as glomerular and tubulointerstitial changes, inflammatory cell infiltration, and fibrosis in kidney biopsy samples, Koç University Pathology Lab , Koç University, Istanbul, Turkey.
2023 – 2024	An investigation into data harmonization tools for organ transplantation, Transplant Immunology Research Center of Excellence Lab (TIREX) , Koç University, Istanbul, Turkey.
2022 – Current	Creating a centralized registry for transplantation data, Transplant Immunology Research Center of Excellence Lab (TIREX) , Koç University, Istanbul, Turkey.
2019 – 2021	Building an AI predictive model to predict the Bio-Impedance Analysis (BIA) phase angle value and extracting the most important determinants of it using adults' body composition data, Nutrition Assessment Lab (NALab) , Tabriz University of Medical Science, Tabriz, Iran.
2020 – 2021	Classification of Phonocardiogram (PCG) and Electrocardiogram (ECG) signals using deep learning, signal processing, and optimization algorithms, Biomedical Engineering Lab (BIOLab) , Tabriz University of Medical Science, Tabriz, Iran.
2020 - 2021	Urinary sediment image classification and segmentation along with image synthesis using Generative Adversarial Networks (GANs), Danesh Pathobiology Laboratory (DPLab) , Tabriz, Iran.
2020	Analyzing body composition data for classifying healthy and depressed obese women using machine learning methods, Nutrition Research Lab (NRLab) , Tabriz University of Medical Science, Tabriz, Iran.
2020	Cluster analysis of claustrophobic patients using hard clustering and fuzzy clustering methods in machine learning, Brain and Cognitive Neuroscience Laboratory (BCNL) , Tabriz University, Tabriz, Iran.



REFERENCES

- **Prof. Dr. Caner Süsal**

Director of Koç University Organ Transplantation Immunology Research Excellence Center (TIREX), Istanbul, Turkey

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- **Assoc. Prof. Dr. Nurcan Tunçbağ**

Department of Chemical and Biological Engineering and the Faculty of Medicine, Koç University, Istanbul, Turkey

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- **Assist. Prof. Beren Semiz**

Department of Electrical and Electronics Engineering, Koç University, Istanbul, Turkey

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